DO YOU KNOW?

Two straight lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ where the coefficients are non-zero, are

(i) parallel if and only if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$; That is, $a_1b_2 - a_2b_1 = 0$

(ii) perpendicular if and only if $a_1a_2 + b_1b_2 = 0$

Progress Check Fill the details in respective boxes

No.	Equations	Parallel or perpendicular	S.No.	Equations	Parallel or perpendicular
1	$5x + 2y + 5 = 0$ $5x + 2y - 3 = 0$		3	$8x - 10y + 11 = 0$ $4x - 5y + 16 = 0$	
2	$3x - 7y - 6 = 0$ $7x + 3y + 8 = 0$		4	$2y - 9x - 7 = 0$ $27y + 6x - 21 = 0$	

5.6.3 Slope of a straight line

The general form of the equation of a straight line is $ax + by + c = 0$. (at least one of a, b is non-zero)

coefficient of $x = a$, coefficient of $y = b$, constant term $= c$.

The above equation can be rewritten as $by = -ax - c$

gives $y = -\frac{a}{b}x - \frac{c}{b}$, if $b \neq 0$ ...(1)

comparing (1) with the form $y = mx + l$

We get, slope $m = -\frac{a}{b}$

$$m = \frac{-\text{coefficient of } x}{\text{coefficient of } y}$$

y intercept $l = -\frac{c}{b}$

$$y \text{ intercept} = \frac{-\text{constant term}}{\text{coefficient of } y}$$

Thinking Corner

How many straight lines do you have with slope 1?

Example 5.30 Find the slope of the straight line $6x + 8y + 7 = 0$.

Solution Given $6x + 8y + 7 = 0$

$$\text{slope } m = \frac{-\text{coefficient of } x}{\text{coefficient of } y} = -\frac{6}{8} = -\frac{3}{4}$$

Therefore, the slope of the straight line is $-\frac{3}{4}$.

Example 5.31 Find the slope of the line which is

(i) parallel to $3x - 7y = 11$ (ii) perpendicular to $2x - 3y + 8 = 0$

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